

5. Discrete random variables		
5.1	The concept of a discrete random variable.	
5.2	The probability function and the cumulative distribution function for a discrete random variable.	Simple uses of the probability function $p(x)$ where $p(x) = P(X = x)$. Use of the cumulative distribution function: $F(x_0) = P(X \leq x_0) = \sum_{x \leq x_0} p(x).$
5.3	Mean and variance of a discrete random variable.	Use of $E(X)$, $E(X^2)$ for calculating the variance of X. Knowledge and use of $E(aX + b) = aE(X) + b,$ $\text{Var}(aX + b) = a^2 \text{Var}(X).$
5.4	The discrete uniform distribution.	The mean and variance of this distribution.